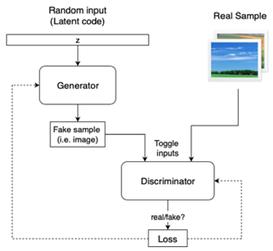
Implementing a Progressive Multi-Scale Design Principle On A Style-Based Architecture for GAN

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Introduction

General Adversarial Networks (GAN) is a relatively new concept in machine learning and is primarily used to produce artificial images indistinguishable from the real ones. Examples of GAN network are, using it to add objects to an image while maintaining the integrity of the image, creating real images from sketches or outlines etc.



Our project creates artificial faces using an adapted GAN network. Creation of artificial faces can be useful for a variety of applications, especially in the creative field. However, a degree of control will explicably increase its use. Our project uses two images, a source and a destination image to create the artificial faces. Features such as the age, gender and pose are determined by the source image, while the destination image determines other features such as skin colour, highlights, hair colours etc. This provides a great degree of control when generating artificial faces, as we can predetermine the features, we want our artificial face to have. For our project we show two levels of style mixing, a coarse and fine level.

An ordinary GAN network used to implement this feature has proven to be not very efficient. We implement a unique GAN network called Style GAN which has produces better results with higher resolutions and computes faster.

However, implementation of these algorithms is extremely costly when considering the implementation time, storage space and computational capacity. This limits the usage of these novel methods to an ordinary user, making the user rely on pretrained weights and giving them limited capacity for modification. To put to perspective, the dataset Flicker Faces HQ (FFHQ) has 70,000 images of the resolution 1024x1024 pixels. The training time of Style GAN on this dataset on a GPU TeslaV100 is estimated to be 41 days!

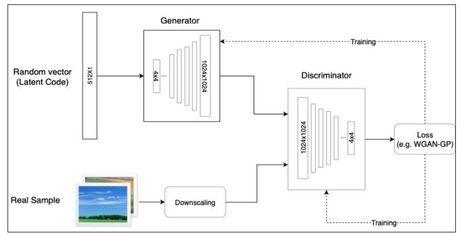
Hence, we use another technique, a fully Progressive Approach to Single-Image Super-Resolution (ProSRs). ProSR is a Single Image Super-Resolution (SISR) method designed upon the principle of multi-scale progressiveness. ProSR can upscale an image up to eight times its original resolution because of the progressive approach. The model will train in a such a manner that it will first focus on improving its results for the lower resolutions and then gradually increasing the scale, making its training harder with each progressive level. Progressive approaches reduce losses significantly while training a model compared to other enhancement methods [3]. To conclude, we trained a model using Style GAN on a training dataset with a resolution of 128x128 pixels and the results produced by this model were enhanced using ProSRs to obtain an image of size 1024x1024.

Methods

StyleGAN

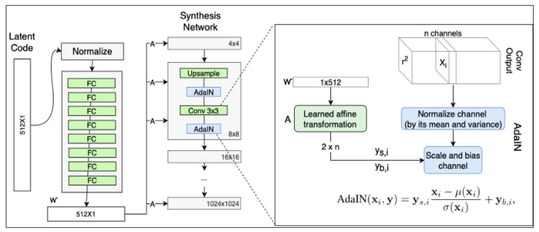
An adapted GAN approach if Progressive GAN (or Pro GAN). This particular network serves as the main principle for our two models. ProGAN trains the network on multiple resolutions, increasing the resolution of training only after a certain decrease in its loss. It is very efficient in training the weights because of the increasing difficulty level of training. As the resolution increases the computational time also increases.

Figure-1: ProGAN Architecture



StyleGAN has adapted ProGAN and utilizes the fact that every layer of training, consists of information for a particular resolution. Traditionally, the weights are fed to the first layer, and then transferred to the others, StyleGAN however feeds the weights to each layer, giving us results obtained by each layer, which help us for style-mixing. During style mixing, two artificial faces are generated, and due to StyleGAN we have access to the information of their layers. Depending on our style level, coarse or fine, the corresponding information of the layers is extracted from the two images and merged to create a new artificial face according to our specifications.

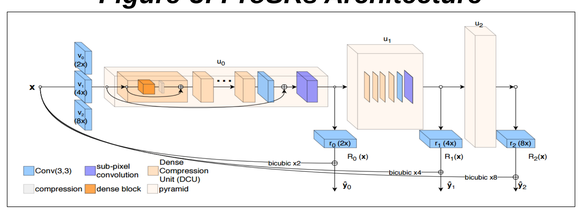
Figure-2: StyleGAN Architecture



ProSRs

ProSRs network reconstructs a high resolution image in intermediate steps by progressively performing a 2× upsampling of the input from the previous level.

Figure-3: ProSRs Architecture



Results

The Style GAN was trained on 1030 images of the resolution 128x128 belonging to the FFHQ dataset. The model is still currently being trained and approximately shows a reduction of 7 points in FID50k every five hours. The model is being trained on the GPU Tesla V100. The models training is still currently in progress as FID50k results have not shown convergence.

The ProSRs was trained using 1030 low resolution and high resolution images. The model was trained on the GPU GeForce GTX 1050. The model took approximately 24 hours to train, but reached convergence within the first few epochs itself. It takes the model approximately 0.5 seconds to generate the high resolution image.

Figure-4: ProSRs x8 Upscaling

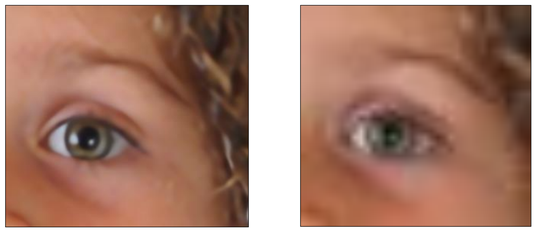


Figure-5: Artificial Faces

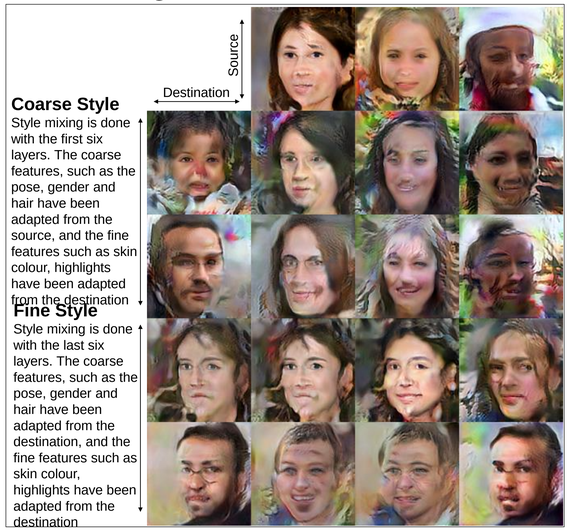
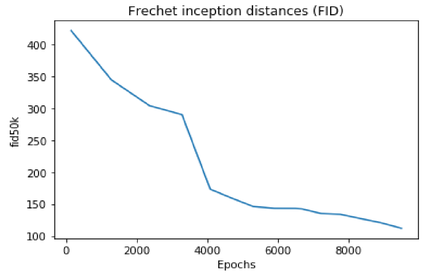


Figure-6: Fid 50k for Style GAN



Conclusions

As evident from the posters, the model still needs significant amount of training, which is currently still in progress. [2] estimated the training period for 70,000 datasets of 256 resolution on a single GPU to be 14days and 22 hours. Our estimate for training our model only on 1030 datasets of 128 resolution is around 7days, hence we have our suspicion that the estimates given by the paper are biased and the actual training period will take significantly more time. The original paper trained their model on 8 Tesla V100 GPUS and took around a week of training. Comparatively, our model has shown tremendous results for the significantly less computational power. The current FID50k score is around 110 which is why the artificial faces being generated are still distorted to a great degree, but with each reduction of the FID50k score, the model has shown significant improvement in generating the artificial faces. Ideally, a score below 10 would reap faces that nearly indistinguishable from a real face, however we estimate that a score below 50 would yield conclusive results to observe our style mixing method.

The ProSRs is a benchmark in Single Image Super Resolution methods, because of its significant reduction in training time and fast computation.. The images produced by Style GAN are appropriately enhanced by the ProSRs model, but due to the current distortion in the images, its enhancement can not truly be appreciated.

The drawbacks however, is that regardless of enhancement, information is still lost. Fine super resolution details such as baby hairs, freckles, facial hair, and texture can not be produced by the ProSRs as its enhancement gives an airbrushed feel to the original image. Hence, the artificial faces reproduced can either be used in their original resolution, or be enhanced if the application does not require a realistic reproduction of the faces. This issue can further be resolved by adding noises to each of the layers of the Style Gan network just like what is done for the weights. The concept behind this is that fine details, such as uneven coloring of the skin to a great extent can be considered as noise itself. Addition of noise can help reduce the airbrushed look that is produced by the enhancement. However, a separate model will have to created for addition of noise after enhancement which does not serve our purpose of using low resolution for the ease of computation, therefore it will be more efficient to simply train the model with a high resolution dataset itself.

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